Colored noises

Random signals or processes with specific spectral characteristics, often described in terms of their power spectral density (PSD)

1. **White Noise**
   - **Definition**: equal power at every frequency.
   - **PSD**: $S(f) \propto$ constant or $S(f) \propto f^0$
   - **Characteristics**: uncorrelated in time
2. **Pink Noise**
   - **Definition**: also known as $1/f$ noise, each octave (halving of frequency) carries an equal amount of power.
   - **PSD**: $S(f) \propto \frac{1}{f}$
3. **Brown Noise (Red Noise)**
   - **Definition**: also known as Brownian noise or red noise
   - **PSD**: $S(f) \propto \frac{1}{f^2}$
4. **Blue Noise**
   - **PSD**: $S(f) \propto f$
   - **Characteristics**: less common in practical applications.
5. **Violet Noise**
   - **Definition**: aka purple noise
   - **PSD**: $S(f) \propto f^2$
6. **Gray Noise**
   - **Definition**: defined perceptually, with the intention of sounding equally loud at all frequencies to the human ear, correcting for the ear's sensitivity to different frequencies.
   - **PSD**: Adjusted based on human hearing sensitivity curves.
   - **Characteristics**: It aims to provide a flat perceived sound spectrum, unlike white noise which is flat in power but not in perceived loudness.
   - **Sound**: often used in psychoacoustic testing.
7. **Green Noise**
   - **Definition**: a variant of pink noise with a peak in the middle frequencies, roughly corresponding to the peak sensitivity of human hearing.
   - **PSD**: Emphasizes frequencies around 500 Hz to 2000 Hz.
   - **Sound**: It is used for relaxation and sleep aids, sounding like the background noise of a forest or ocean waves.

Noise Type	Power Spectral Density (PSD)	Sound Characteristics	Common Uses
White Noise	$S(f) \propto f^0$	Hiss, static	Audio testing, sleep aids
Pink Noise	$S(f) \propto \frac{1}{f}$	Balanced, deeper	Music production, sound engineering
Brown Noise	$S(f) \propto \frac{1}{f^2}$	Deep rumbling	Relaxation, sleep aids
Blue Noise	$S(f) \propto f$	High-pitched hiss	Technical applications
Violet Noise	$S(f) \propto f^2$	Sharp, very high-pitched	Technical applications
Gray Noise	Adjusted for human hearing	Perceptually flat	Psychoacoustic testing
Green Noise	Emphasizes mid frequencies	Natural, soothing	Relaxation, nature sounds

Each type of colored noise serves different purposes and is used in various applications based on its spectral characteristics.

Generation by...

- **White noise**: ... regular random number generator;
- **Brown noise**: ... cumulative sum of white noise;
- **Other noises**: ... inverse Fourier of the prescribed PSD of the form $1/f^{\alpha} \times$noise.

Dynamic system:

$$\dot{\boldsymbol{X}} = \boldsymbol{\Phi}(\boldsymbol{X}, t) \ , \quad \boldsymbol{X} : \mathbb{R} \to \mathbb{R}^d \ , \boldsymbol{\Phi} : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^d$$

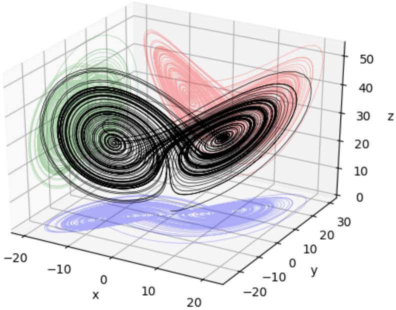
Discrete version:

$$\boldsymbol{X}_{n+1} = \boldsymbol{\Phi}(\boldsymbol{X}_n, n)$$

Example: Lorenz system

$$\begin{cases} \dfrac{\mathrm{d}x}{\mathrm{d}t} = \sigma(y - x), \\[2ex] \dfrac{\mathrm{d}y}{\mathrm{d}t} = x(\rho - z) - y, \\[2ex] \dfrac{\mathrm{d}z}{\mathrm{d}t} = xy - \beta z. \end{cases}$$

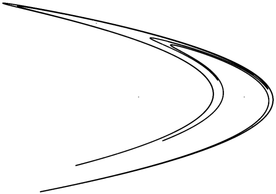
Chaotic for $\rho = 28$, $\sigma = 10$, $\beta = 8/3$.
Applications: atmospheric phenomena (x, y, z velocity, horizontal and vertical temperature gradient), lasers (intensity, polarization, population inversion), thermosyphon, DC engine, stb.



Example: Henon map

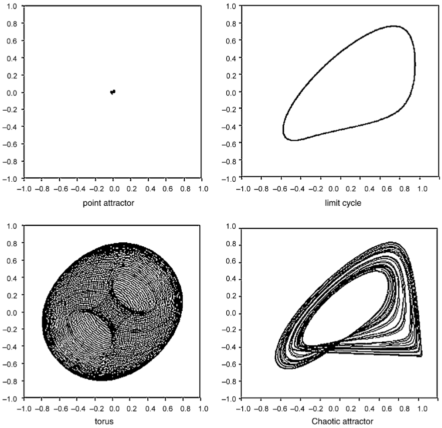
$$\begin{cases} x_{n+1} = 1 - ax_n^2 + y_n \\ y_{n+1} = bx_n\,. \end{cases}$$

Chaotic for $a = 1.4$, $b = 3.0$.



Types of attractors:

- point
- limit cycle
- torus
- chaotic (strange)



$$\boldsymbol{X}(t) \quad \xrightarrow{\text{observation}} \quad x(t) \equiv g(\boldsymbol{X}(t))$$

original d dimensional dynamics — univariate time series

Time series = observation = low dimensional projection of the dynamics.

How to acquire information about the dynamics?

Steps:

- reconstruct the dynamics in the phase space
- characterize the reconstructed attractor
- test/validate using surrogate data

Takens' theorem

The dynamics with an attractor of dimension d_A can be reconstructed from an observation via function g in a Euclidean space of dimension $m > 2d_A$.

$$\boldsymbol{X}(t) \xrightarrow{\text{observation}} x(t) \equiv g(\boldsymbol{X}(t)) \xrightarrow{\text{reconstruction}} \boldsymbol{X}_i = [x_i, x_{i+l}, x_{i+2l}, \ldots, x_{i+(m-1)l}]$$

original d dimensional dynamics — univariate time series — m dimensional reconstructed dynamics

Multivariate time series:

$$\boldsymbol{X}(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$$

Discrete time: $t_n = n\Delta t$

$$\boldsymbol{X}_n = \begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix}, \quad n = 0, 1, 2, \ldots \quad \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \cdots\cdots \begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix}$$

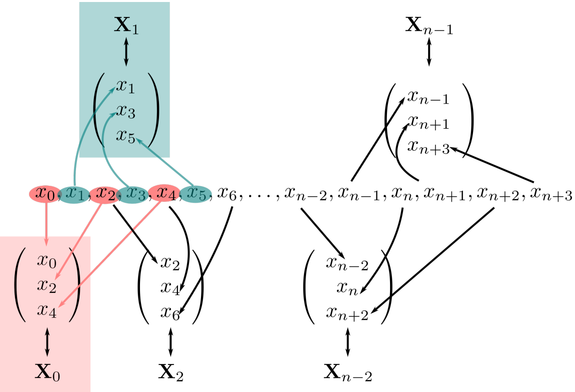
$$\updownarrow \quad \updownarrow \quad \updownarrow \quad \updownarrow$$

$$\mathbf{X}_0 \quad \mathbf{X}_1 \quad \mathbf{X}_2 \quad \mathbf{X}_n$$

Nonlinear methods: time delay embedding

Univariate time series $x(t)$:
Discrete time: $t_n = n\Delta t$

x_n , $n = 0, 1, 2, \ldots$



Parameters: m - number of dimensions (here 3), l - delay (here 2)

Nonlinear methods: time delay embedding

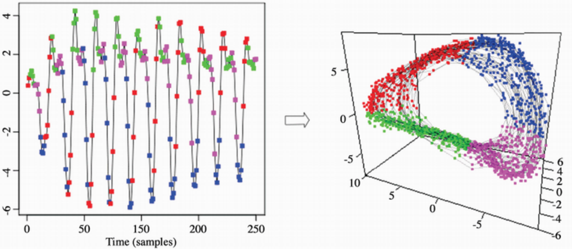
Generally:

- Embedding the **continuous** time series $x(t)$ into m dimensions with delay τ:

$$\boldsymbol{X}(t) = [x(t), x(t+\tau), \ldots, x(t+(m-1)\tau)]$$

- Embedding the **discrete** time series $x(t)$ into m dimensions with delay l:

$$\boldsymbol{X}_n = \left[x_n, x_{n+l}, \ldots, x_{n+(m-1)l}\right]$$



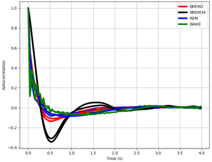
Example: $m = 3$, $l = 11$.

D. Precup and S. Mannor and J. Pineau and J. Frank : *Time Series Analysis Using Geometric Template Matching*, IEEE Transactions on Pattern Analysis & Machine Intelligence, **35** (2013)

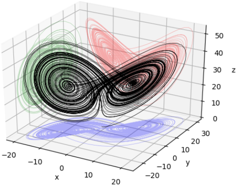
How to set the values of m and τ (l)?

How to set the values of m and τ (l)?

- $m \cdot \tau \sim$ the first null point of the autocorrelation function



- m dimensions $\leftarrow$ false neighbor method.



Nonlinear methods: dynamic properties

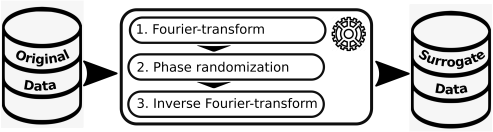
Methods, measures

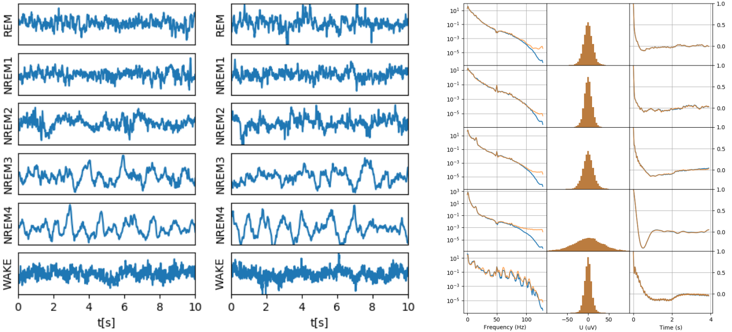
- nonlinear forecast;
- testing determinism;
- testing reversebility;
- characterize chaoticity;
- other measures.

- How do we know that the phenomenon (producing the time series) is indeed of nonlinear nature?
- Create surrogate data featuring the original data's linear measures.
- The deviation in the nonlinear measures between the original data and the surrogate data characterize the nonlinear properties of the former.

How to proceed?

Nonlinear methods: surrogate data

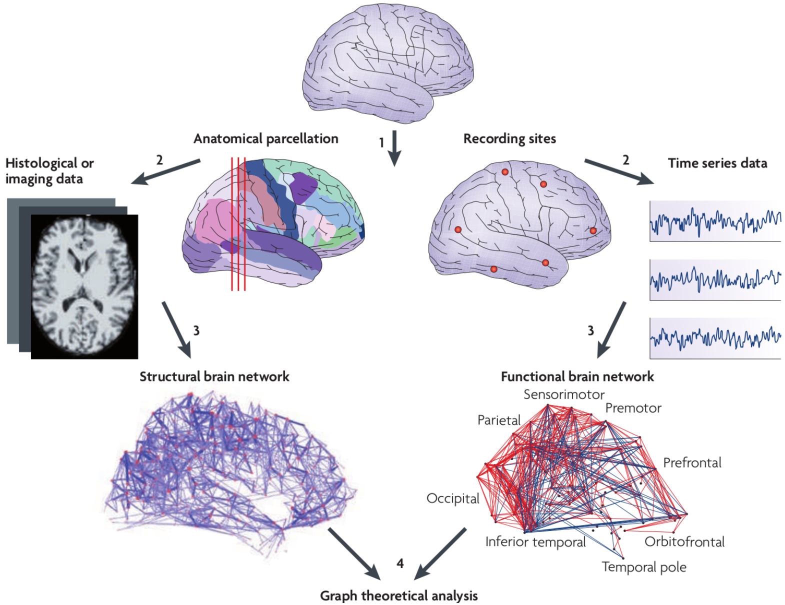




EEG Surrogate data Spectrum Distribution Autocorr.

- As the power spectral density is the Fourier transform of the autocorrelation function, the latter is automatically reproduced.
- The distribution of the surrogate data differs (closer to normal)
- There are more sophisticated methods to reproduce the distribution as well

Connectivity measures

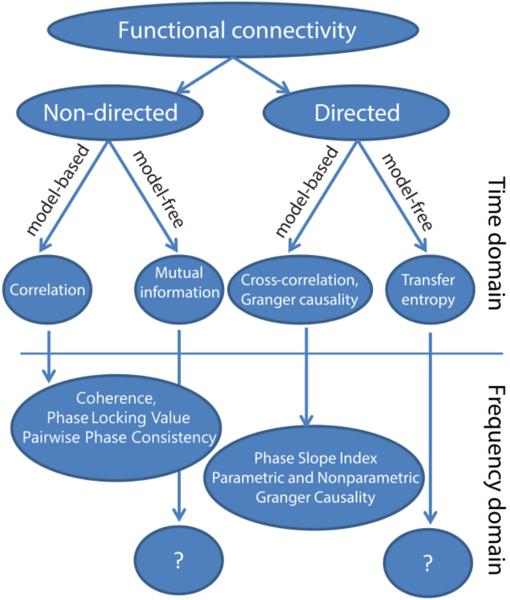

Anatomical parcellation
1
Recording sites
2
Histological or imaging data
2
Time series data
3
3
Structural brain network
Functional brain network
Sensorimotor
Premotor
Parietal
Prefrontal
Occipital
Inferior temporal
Orbitofrontal
Temporal pole
4
Graph theoretical analysis

Classifications

- local
- global

- undirected
- directed

- linear
- non-linear



A.M. Bastos and J.M. Schoffelen, *A Tutorial Review of Functional Connectivity Analysis Methods and Their Interpretational Pitfalls*, Front. in Systems Neurosci., **9** 175 (2016)

List of connectivity measures

- correlation
- coherence
- imaginary coherence
- phase lag index
- signed phase lag index
- weighted phase lag index
- directed partial coherence
- scaled correlation
- mutual information
- Granger causality
- synchronization likelihood
- partial directed coherence
- transfer entropy
- directed transfer function
- phase transfer entropy

Covariance and correlation

Random variables: x, y
Expectance value: $\langle x \rangle$
Variance: $\sigma_x^2 = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2$
Covariance:

$$c_{xy} = \langle (x - \langle x \rangle)(y - \langle y \rangle) \rangle = \langle xy \rangle - \langle x \rangle \langle y \rangle$$

If $x \equiv \pm y \rightarrow c_{xy} = \pm \sigma_x^2$
If x independent of $y \rightarrow c_{xy} = 0$
If $\langle x \rangle = \langle y \rangle = 0$

$$c_{xy} = \langle xy \rangle$$

(Pearson's) correlation coefficient:

$$r_{xy} = \frac{c_{xy}}{\sigma_x \sigma_y} \in [-1, +1]$$

For $x_1, x_2, \ldots x_n$ samples of x:

$$\langle x \rangle = \frac{1}{n} \sum_{i=1}^{n} x_i \qquad \text{mean value}$$

$$r_{xy} = \frac{\sum_i (x_i - \langle x \rangle)(y_i - \langle y \rangle)}{\sqrt{\sum_i (x_i - \langle x \rangle)^2 \sum_j (y_j - \langle y \rangle)^2}} \qquad \text{uncorrected Pearson's coefficient}$$

Time series: continuous $x(t), t \in \mathbb{R}$, discreet x_i , $i \in \mathbb{Z}$ ordered set

$$\langle x \rangle = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{+T} x(t)dt = 0 \quad \text{(must be!!)}$$

Cross correlation

$$c_{xy}(\tau) = (x\star y)(\tau) = \int\limits_{-\infty}^{\infty} x^*(t)y(t+\tau)dt, \qquad c_{xy}(k) = \frac{1}{n}\sum_i x_i^* y_{i+k} \ , \quad \tau, k\text{: lag}$$

$$\sigma_x^2 = c_{xx}(0) = \int\limits_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{n}\sum_i |x_i|^2$$

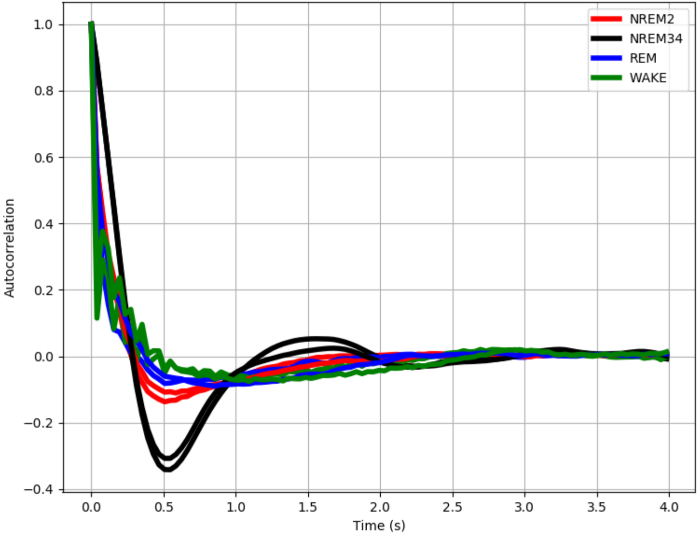
Cross correlation coefficient

$$r_{xy}(\tau) = \frac{c_{xy}(\tau)}{\sigma_x \sigma_y} \ , \qquad r_{xy}(k) = \frac{\sum_i (x_i - \langle x \rangle)(y_{i+k} - \langle y \rangle)}{\sqrt{\sum_i (x_i - \langle x \rangle)^2 \sum_j (y_j - \langle y \rangle)^2}}$$

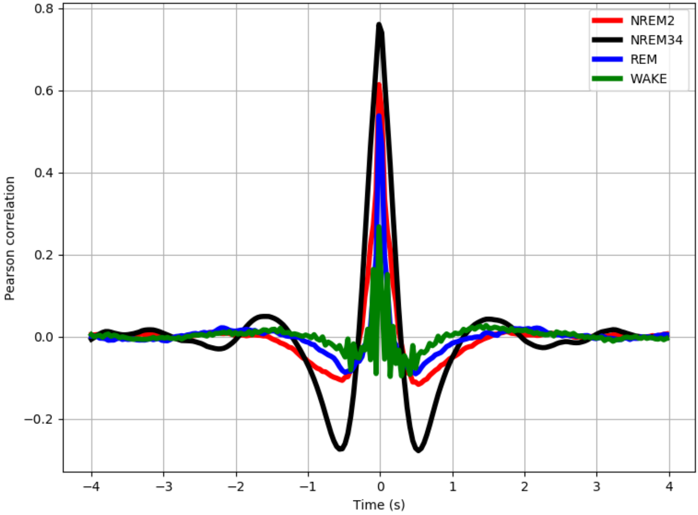
Autocorrelation coefficient

$$r_{xx}(\tau), \qquad r_{xx}(k)$$

So for a given lag, τ, the (normalized) cross correlation coefficient and the uncorrected Pearson's coefficient are calculated in the same way.



NREM2
NREM34
REM
WAKE
Autocorrelation
Time (s)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0



Channels C3 & C4

Phase coherence

Characterizes the stability of the phase differences between the two signals, for each individual frequency component.

Fourier transform

$$\mathscr{F}[x] \equiv X(\nu) \equiv \int_{-\infty}^{+\infty} dt x(t) e^{2\pi i \nu t} \ ,$$

Cross spectral density

$$C_{xy}(\nu) \equiv X^*(\nu) Y(\nu) = \int_{-\infty}^{+\infty} dt \int_{-\infty}^{+\infty} dt' x^*(t) y(t') e^{2\pi i \nu (t'-t)} = \qquad \tau = t' - t$$

$$= \int_{-\infty}^{+\infty} d\tau \int_{-\infty}^{+\infty} dt \ x^*(t) y(t+\tau) e^{2\pi i \nu \tau} =$$

$$= \int_{-\infty}^{+\infty} d\tau \ c_{xy}(\tau) e^{2\pi i \nu \tau} = \mathscr{F}[c_{xy}]$$

Power spectral density

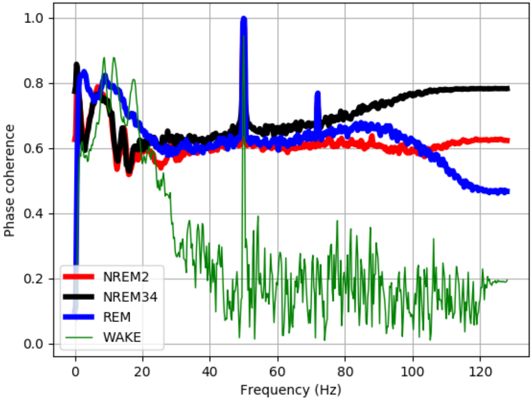
$$P_x(\nu) \equiv C_{xx}(\nu) = |X(\nu)|^2 = \mathscr{F}[c_{xx}] \quad \rightarrow \quad \text{Wiener–Khinchin theorem}$$

$$\int_{-\infty}^{+\infty} P_x(\nu) d\nu = \int_{-\infty}^{+\infty} d\tau \int_{-\infty}^{+\infty} dt \, x^*(t) x(t+\tau) \underbrace{\int_{-\infty}^{+\infty} d\nu \ e^{2\pi i \nu \tau}}_{\delta(\tau)} = \int_{-\infty}^{+\infty} dt |x(t)|^2$$

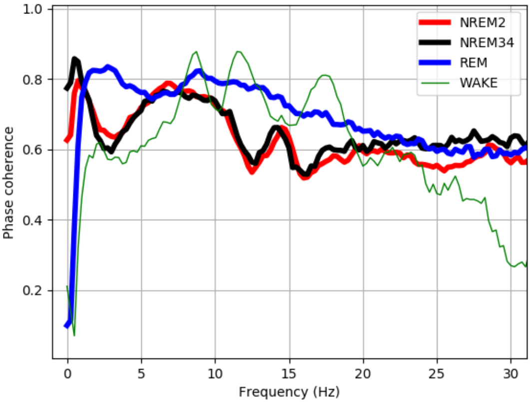
Phase coherence

$$Coh_{xy}(\nu) = \frac{|C_{xy}|}{\sqrt{P_x P_y}} \;\in\; [0,1]$$

scale invariant $(x(t) \to \alpha x(t), y(t) \to \beta y(t))$



Channels C3 & C4



Channels C3 & C4